

Algebraic Reasoning

MA.8.AR.1.1

Benchmark

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Example: The expression $(3x^3y^{-2})^3$ is equivalent to $27x^9y^{-6}$.

Benchmark Clarifications:

Clarification 1: Refer to the K-12 Formulas (Appendix E) for the Laws of Exponents.

Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.3, MA.8.NSO.1.7
- MA.8.AR.1.3

Terms from the K-12 Glossary

- Base
- Expression
- Integers
- Monomial

Vertical Alignment

Previous Benchmarks

- MA.7.NSO.1.1
- MA.7.NSO.2.1

Next Benchmarks

- MA.912.NSO.1.2

Purpose and Instructional Strategies

In Grade 7, students applied the Laws of Exponents to evaluate and generate numerical expressions, limited to whole-number exponents and rational number bases. In Grade 8, students extend their knowledge of the Laws of Exponents to generate equivalent algebraic expressions with integer exponents and monomial bases. In Algebra 1, students will use their knowledge of the Laws of Exponents to generate equivalent algebraic expressions with rational and variable exponents.

- At the onset of learning about exponents, students learn that it is a way to write expanded multiplication in a more condensed form. It is foundational that students understand that the value of the exponent determines how many times the base is multiplied by itself.
- This benchmark can be paired with MA.8.NSO.1.3 which helps students work within numerical expressions using integer exponents and rational bases. Students should move from numerical expressions to algebraic expressions to best enhance their conceptual understanding of the Laws of Exponents.

- A strategy for developing meaning for integer exponents is to make use of patterns as shown below:

Patterns in Exponents	
x^5	$x \cdot x \cdot x \cdot x \cdot x$
x^4	$x \cdot x \cdot x \cdot x$
x^3	$x \cdot x \cdot x$
x^2	$x \cdot x$
x^1	x
x^0	1
x^{-1}	$\frac{1}{x}$
x^{-2}	$\frac{1}{x \cdot x}$
x^{-3}	$\frac{1}{x \cdot x \cdot x}$
x^{-4}	$\frac{1}{x \cdot x \cdot x \cdot x}$
x^{-5}	$\frac{1}{x \cdot x \cdot x \cdot x \cdot x}$

- For conceptual understanding, instruction should focus on one law at a time and students should derive the law of exponents through experience and reason. During instruction, include examples that show the expansion of the bases with the use of the exponents to show equivalence.
- For mastery of this benchmark, monomials can be defined in the following way: a base may be a product of a coefficient and one or more variables with integer exponents. This limitation should not prevent students from understanding that a negative exponent can be represented equivalently as a positive exponent with the reciprocal base (changing numerator to denominator or denominator to numerator).

Common Misconceptions or Errors

- When working with negative exponents, students may not understand the connection to fractions and values in the denominator.
- Students may incorrectly multiply the exponent with the base number.

Strategies to Support Tiered Instruction

- Teachers should write out the expanded form and then use it to model condensed multiplication as exponential form, providing opportunities to notice patterns as discussed in MA.8.NSO.1.3. Teachers can use the “Patterns in Exponents” chart shown in the Purpose and Instructional Strategies section with the right-side blank so that students can begin to complete and understand the patterns of integer exponents.
- Teachers should re-emphasize the structure of exponents, and how they indicate the number of times the base is multiplied by itself as denoted by the exponent.

- Teacher provides a review of the relationship between the base and the exponent by modeling an example of operations using a base and exponent.
 - For example, evaluate 6^3 .

6^3 which is equivalent to $6 \cdot 6 \cdot 6$ which is equivalent to 216.

Instructional Tasks

Instructional Task 1 (MTR.2.1)

Two students were working on generating equivalent expressions for $(15xy^2)^3$, and showed their solutions below.

Rachel's Answer	Justina's Answer
$15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$	$3,375x^3y^6$

The teacher said Rachel and Justina both have the correct answer. Do you agree with the teacher? Explain your reasoning.

Instructional Task 2 (MTR.5.1)

Create a pattern using the expanded form of the base, m , between m^{-5} and m^5 . Explain why m^0 is equal to 1.

Instructional Items

Instructional Item 1

Write x^5x^8 with the variable x used only one time.

Instructional Item 2

An expression is given.

$$\left(\frac{a^3}{4b^{-7}}\right)^5$$

Write an equivalent expression with only two exponents and no negative exponents.

Instructional Item 3

Write $y^{-3}z^{-4}$ with only positive exponents.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*